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United States Patent	12404127
Kind Code	B2
Date of Patent	September 02, 2025
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Image recording apparatus and feed tray

Abstract

An image recording apparatus includes: a feed tray; a feed roller configured to feed a sheet medium accommodated in the feed tray; and a recording unit. The feed tray includes: a first accommodating part configured to accommodate a rolled body formed by rolling the sheet medium; a second accommodating part configured to accommodate a plurality of sheet media in a stacked state, each of the sheet media being the sheet medium; a contact surface configured to contact, from below, the sheet medium unrolled from the rolled body accommodated in the first accommodating part; and a support member having a support surface configured to support the sheet media accommodated in the second accommodating part. The feed roller is movable between a contact position at which the feed roller contacts the contact surface and a separation position at which the feed roller is away from the contact surface.

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Appl. No.:	18/150312
Filed:	January 05, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230242361 A1	Aug. 03, 2023

Foreign Application Priority Data

JP	2022-012863	Jan. 31, 2022
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Publication Classification

Int. Cl.: B65H1/26 (20060101); B65H1/02 (20060101); B65H3/44 (20060101); B65H5/06 (20060101); B65H7/02 (20060101)

U.S. Cl.:

CPC B65H5/062 (20130101); B65H1/027 (20130101); B65H3/44 (20130101); B65H7/02 (20130101); B65H2301/5111 (20130101); B65H2402/30 (20130101)

Field of Classification Search

CPC: B65H (1/266); B65H (3/44); B65H (3/0684); B65H (16/08); B65H (1/04); B65H (2405/33); B65H (2405/331); B65H (2405/34); B65H (2405/40); B41J (11/58); B41J (11/48); B41J (11/485); B41J (15/044); B41J (11/50); G03G (15/6517)

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Background/Summary

REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority from Japanese Patent Application No. 2022-012863 filed on Jan. 31, 2022. The entire content of the priority application is incorporated herein by reference.

BACKGROUND ART

(2) Conventionally, there is known a facsimile machine (image recording apparatus) having a paper feed cassette (feed tray) capable of accommodating a rolled paper (rolled body) and cut papers (sheet media). This facsimile machine has a conveyance roller for feeding the rolled paper accommodated in the paper feed cassette from the paper feed cassette and a pick-up roller for feeding a sheet of cut paper accommodated in the paper feed cassette from the paper feed cassette.

DESCRIPTION

- (3) In the facsimile machine described above, the rolled paper and the cut papers accommodated in the paper feed cassette are fed from the paper feed cassette by separate rollers. In other words, the rolled paper and the cut papers accommodated in the paper feed cassette are fed from the paper feed cassette by separate paths. Therefore, the apparatus becomes larger.
- (4) An object of the present teaching is to provide an image recording apparatus and a feed tray capable of reducing the size of the apparatus.
- (5) According to a first aspect of the present teaching, there is provided an image recording apparatus, including: a feed tray; a feed roller configured to feed a sheet medium accommodated in the feed tray; and a recording unit configured to record an image on the sheet medium fed by the feed roller, wherein the feed tray includes: a first accommodating part configured to accommodate a rolled body formed by rolling the sheet medium; a second accommodating part configured to accommodate a plurality of sheet media in a stacked state, each of the sheet media being the sheet medium; a contact surface configured to contact, from below, the sheet medium unrolled from the rolled body accommodated in the first accommodating part; and a support member having a support surface configured to support the sheet media accommodated in the second accommodating part, the feed roller is configured to be movable between a contact position at which the feed roller contacts the contact surface and a separation position at which the feed roller is away from the contact surface, and the support member is configured to be movable between a first position at which the support member is outside a movement path of the feed roller and a second position at which the support member is on the movement path of the feed roller with the support surface facing upward.
- (6) According to a second aspect of the present teaching, there is provided a feed tray, including; a first accommodating part configured to accommodate a rolled body formed by rolling the sheet medium; a second accommodating part configured to accommodate a plurality of sheet media in a stacked state, each of the sheet media being the sheet medium; a roller configured to contact, from below, the sheet medium unrolled from the rolled body accommodated in the first accommodating part; and a support member having a support surface configured to support the sheet media accommodated in the second accommodating part, wherein the support member is configured to be movable between a cover position at which the support member covers the roller with the support surface facing upward and a separation position at which the support member is farther from the roller as compared with the cover position.
- (7) According to the image recording apparatus and feed tray of the present teaching, when feeding the sheet medium in the first accommodating part from the feed tray, the support member is set to the first position, so that the sheet medium in contact with the contact surface can be fed from the feed tray by the feed roller. When feeding the sheet medium in the second accommodating part from the feed tray, the supporting member is set to the second position, so that the sheet medium supported on the support surface can be fed from the feed tray by the feed roller. In other words, one feed roller can feed the sheet medium in the first accommodating part and the sheet medium in the second accommodating part from the feed tray along the same path. Therefore, the size of the apparatus can be reduced.

Description

- (1) FIG. 1 is a schematic side view depicting an internal structure of a printer according to an embodiment of the teaching.
- (2) FIG. 2 is a perspective view of a feed tray depicted in FIG. 1, with a flap in a first position.
- (3) FIG. 3 is a perspective view of the feed tray depicted in FIG. 1, with the flap in a second position.
- (4) FIG. 4A is a cross-sectional view along a IVA-IVA line of the feed tray depicted in FIG. 2, and

FIG. 4B is a cross-sectional view along a IVB-IVB line of the feed tray depicted in FIG. 3.

(5) FIG. 5 is an enlarged view of an area near the flap depicted in FIG. 3.

(6) FIG. 6 is an enlarged view of an area near a protrusion in a link member when the flap is in the first position.

(7) FIG. 7 is a cross-sectional view along a VII-VII line of the feed tray depicted in FIG. 3.

(8) FIGS. 8A and 8B depict a support plate and a display on the flap, FIG. 8A depicts the flap in the first position, and FIG. 8B depicts the flap in the second position.

(9) FIG. 9 is an enlarged view of a rolled paper setting procedure label and a cut paper setting procedure label.

(10) FIGS. 10A and 10B illustrate configurations of a paper sheet presence/absence detecting unit, FIG. 10A depicts a state where no paper sheet is detected, and FIG. 10B depicts a state where a paper sheet is detected.

(11) FIG. 11 depicts a state where an arm supporting a feed roller is in a retracted position.

(12) FIGS. 12A and 12B depict a feed tray according to a modification of the embodiment.

(13) A printer **100** (image recording apparatus of the present teaching) according to one embodiment of the present teaching is described below with reference to FIG. 1. Up-down, front-rear, and left-right directions in FIG. 1 are defined as up-down, front-rear, and left-right directions of the printer **100**, respectively.

(14) (Overall Configuration of Printer **100**)

(15) The printer **100** mainly includes a housing **100a**, a feed mechanism **3**, a conveyance mechanism **4**, a cutter **5**, a head **6** (corresponding to the “recording unit” of the present teaching), a paper discharge tray **7**, a paper sheet presence/absence detecting unit **20** (see FIG. 10) and a controller **10**.

(16) The feed mechanism **3** has a feed tray **1**, a feed roller **2**, and a separation strip **31**. The feed tray **1** is located below the head **6** in a housing **100a**. An opening **101** is formed in a front wall of the housing **100a**. The feed tray **1** can be inserted into and removed from the housing **100a** along the front-rear direction through the opening **101**. In other words, the feed tray **1** can be moved along the front-rear direction between inside the housing **100a** and outside the housing **100a**.

(17) The feed tray **1** can accommodate a rolled body R and cut papers Kp. The feed tray **1** has a first accommodating part **8** that can accommodate the rolled body R and a second accommodating part **9** that can accommodate a plurality of cut papers Kp (an example of the sheet media) in a stacked state. The rolled body R has a cylindrical core member Rc, and a rolled paper Rp (an example of the sheet medium) is rolled around an outer circumference of the cylindrical core member Rc. A width direction of the rolled paper Rp unrolled from the rolled body R in the feed tray **1** and a width direction of the cut papers Kp in the feed tray **1** coincide with the left-right direction.

(18) In FIG. 1, both the rolled body R and the cut papers Kp are accommodated in the feed tray **1** for illustration. However, when recording images with the printer **100**, any one of the rolled body R and the cut papers Kp is accommodated in the feed tray **1**.

(19) The feed roller **2** feeds the rolled paper Rp unrolled from the rolled body R accommodated in the first accommodating part **8** or a sheet of the cut papers Kp accommodated in the second accommodating part **9**, from the feed tray **1**. The rolled paper Rp and the cut paper Kp fed by the feed roller **2** are fed from feed tray **1** along the same path. In the following description, when the cut paper Kp and the rolled paper Rp are not distinguished, they are referred to as a “paper sheet P”.

(20) The feed roller **2** is configured with two rollers **2a** supported on a tip end of an arm **2b**. The two rollers **2a** are arranged side by side along the left-right direction with the arm **2b** intervening therebetween. The two rollers **2a** are spaced apart from each other. The two rollers **2a** rotate around an axis along the left-right direction by driving a feed motor not depicted in the figure.

(21) The arm **2b** is rotatably supported by a spindle **2c**. The spindle **2c** is supported by the housing

100a. The arm **2b** is biased by an undepicted biasing member so that the feed roller **2** approaches a bottom surface **11a** of the feed tray **1**. The feed roller **2** is movable between a position at which the feed roller **2** contacts rollers **18** (see FIG. **2**) arranged on the bottom surface **11a** of the feed tray **1** and a position at which the feed roller **2** is away from the rollers **18**, by a rotation of the arm **2b** around the spindle **2c**. As will be described in more detail later, the arm **2b** can be retracted to a retracted position where the arm **2b** does not interfere with the body portion **16** of the feed tray **1**, in conjunction with a movement of the feed tray **1** from an inside of the housing **100a** to an outside of the housing **100a**.

(22) When the feed motor is driven by the controller **10**, the feed roller **2** rotates to apply conveying force in a direction from front to rear to the paper sheet **P** in contact with the feed roller **2**. This causes the paper sheet **P** to be fed from the feed tray **1**.

(23) The separation strip **31** is provided to prevent feeding the cut papers **Kp** in a state of being overlapped with each other when feeding the cut papers **Kp** from the feed tray **1**. The separation strip **31** is located downstream of the feed tray **1** in a feed direction (from front to rear; referred to simply as a “feed direction” in the following description).

(24) The separation strip **31** is inclined so that a rear end is positioned higher than a front end. When the feed tray **1** is installed in the housing **100a** and is inside the housing **100a**, the separation strip **31** is positioned in a center of the feed tray **1** in the left-right direction. The separation strip **31** contacts a central part of the cut paper **Kp** in the width direction, and separates the cut paper **Kp** in contact with the feed roller **2** from the other sheets of the cut papers **Kp** in a stacked state. The paper sheet **P** fed from the feed tray **1** contacts the separation strip **31** and then is guided diagonally upward.

(25) The conveyance mechanism **4** has an intermediate roller pair **41**, a conveyance roller pair **42**, a paper discharge roller pair **43**, and a guide member **44**. The intermediate roller pair **41** is configured by a drive roller which is rotated by driving of an undepicted intermediate motor, and a driven roller which is rotated along with the rotation of the drive roller. When the intermediate motor is driven by a control of the controller **10**, the intermediate roller pair **41** rotates while nipping the paper sheet **P** to convey the paper sheet **P**. The intermediate roller pair **41** is located above the separation strip **31**. The intermediate roller pair **41** conveys the paper sheet **P** which is fed from the feed tray **1** by the feed roller **2** and then guided diagonally upward by the separation strip **31** while nipping the paper sheet **P**. The guide member **44** is located above the intermediate roller pair **41**. The guide member **44** guides the paper sheet **P** forward, after the paper sheet **P** has been conveyed upward by the intermediate roller pair **41**.

(26) The conveyance roller pair **42** is configured by a drive roller which is rotated by driving of an undepicted conveyance motor and a driven roller which is rotated along with the rotation of the drive roller. The paper discharge roller pair **43** is configured by a drive roller which is rotated by driving of an undepicted paper discharge motor and a driven roller which is rotated along with the rotation of the drive roller. When the conveyance motor and the paper discharge motor, which are not depicted in the figure, are driven by a control of the controller **10**, the conveyance roller pair **42** and the paper discharge roller pair **43** rotate while nipping the paper sheet **P** to convey the paper sheet **P**. The conveyance roller pair **42** is located behind the head **6**, and the paper discharge roller pair **43** is located in front of the head **6**. The conveyance roller pair **42** conveys the paper sheet **P** forward while nipping the paper sheet **P**, after the paper sheet **P** has been guided forward by the guide member **44**. The paper discharge roller pair **43** conveys the paper sheet **P** forward while nipping the paper sheet **P**, after the paper sheet **P** has been conveyed forward by the conveyance roller pair **42**.

(27) The cutter **5** is located between the separation strip **31** and the intermediate roller pair **41**. The cutter **5** is configured, for example, by a disc-shaped rotating blade and a driven blade. The cutter **5** may also be configured by a rotating blade and a fixed blade. The rotating blade of the cutter **5** is rotated by driving of an undepicted cutting motor, and the cutter **5** reciprocates along the left-right

direction. The rolled paper Rp unrolled from the rolled body R and conveyed is cut by the cutter 5 in the width direction of the rolled paper Rp by driving of the cutting motor under a control of the controller 10.

(28) The head 6 includes a plurality of nozzles (not depicted) formed on a lower surface thereof and a driver IC (not depicted). The head 6 records an image on the paper sheet P, which is fed by the feed roller 2 from the feed tray 1, and conveyed by the conveyance mechanism 4. When the driver IC is driven by the controller 10, ink is ejected from the nozzles. When the paper sheet P conveyed by the conveyance mechanism 4 passes through an image recording position facing the lower surface of the head 6, the ink is ejected from the nozzles to form the image on the paper sheet P. The head 6 can be either a line type which ejects the ink from the nozzles in a state of the head 6 being fixed to a predetermined position or a serial type which ejects the ink from the nozzles while moving in the left-right direction.

(29) The paper discharge tray 7 is located in front of the head 6 and above the feed tray 1 in the housing 100a. An opening 102 is formed in the front wall of the housing 100a. The paper discharge tray 7 can be inserted into and removed from the housing 100a along the front-rear direction through the opening 102. The paper discharge tray 7 receives the paper sheet P having the image formed by the head 6 and conveyed forward by the paper discharge roller pair 43.

(30) The paper sheet presence/absence detecting unit 20 detects whether or not the paper sheet P is placed at a feed position where the feed roller 2 can feed the paper sheet P. In other words, the paper sheet presence/absence detecting unit 20 corresponds to the “detecting unit” of the present teaching. The paper sheet presence/absence detecting unit 20 is attached to the arm 2b, as depicted in FIG. 10. A specific configuration of the paper sheet presence/absence detecting unit 20 is described below.

(31) The controller 10 is connected via an internal bus (not depicted) to the feed motor, the intermediate motor, the conveyance motor, the paper discharge motor, the cutting motor, the paper sheet presence/absence detecting unit 20, and the driver IC. The controller 10 includes a CPU (Central Processing Unit), a ROM (Read Only Memory) and a RAM (Random Access Memory). The ROM stores programs and data for the CPU to perform various controls. The RAM temporarily stores data used by the CPU to execute the programs.

(32) In a case that the paper presence/absence detecting unit 20 detects that the paper sheet P is placed in the feed position, the controller 10 controls feed roller 2 to feed the paper sheet P from the feed tray 1.

(33) (Configuration of Feed Tray 1)

(34) Next, with further reference to FIGS. 2 to 8, the configuration of the feed tray 1 will be explained. In the following description, directions of each part of the feed tray 1 will be explained based on a posture of the feed tray 1 in a state where the feed tray 1 is installed in the housing 100a.

(35) The feed tray 1 has a main body 16. As depicted in FIGS. 2 and 3, the main body 16 has a bottom wall 11 and side walls 12 to 15 at edges of the bottom wall 11, and is formed in a box shape with an upward opening.

(36) The bottom wall 11 extends along a plane perpendicular to the up-down direction. An upper surface of the bottom wall 11 is the bottom surface 11a of the feed tray 1. The bottom surface 11a is orthogonal to the up-down direction. The side walls 12 and 13 extend upward from left and right end edges of the bottom wall 11. The side walls 12 and 13 also extend along the front-rear direction from a front end to a rear end of the bottom wall 11a. Notches 12a and 13a are formed at upper ends of the side walls 12 and 13, respectively. The side wall 14 extends upward from a front end edge of bottom wall 11. The side wall 14 extends along the left-right direction from a right end to a left end of the bottom wall 11.

(37) Two side walls 15 are provided at a rear end edge of the bottom wall 11. The two side walls 15 are located at both ends in the left-right direction of the rear end edge of the bottom wall 11. Each

of the sidewalls **15** has a guide surface **15a** which is connected to a rear end of the bottom surface **11a** and is inclined such that an upper end of the guide surface **15a** is located behind a lower end of the guide surface **15a**. When the feed tray **1** is installed in the housing **100a**, the separation strip **31** (see FIG. **1**) is positioned between the two side walls **15** arranged side by side in the left-right direction. The guide surface **15a** of each of the side walls **15**, together with the separation strip **31**, guides the paper sheet P diagonally upward after the paper sheet P has been fed rearward by the feed roller **2**.

(38) The feed tray **1** has a pair of side guides **17** located inside the side walls **12** and **13**. The side guides **17** are apart from each other in the left-right direction. The side guides **17** are supported by the bottom wall **11**, such that the side guides **17** are movable in the left-right direction. The side guides **17** are arranged to face each other in the left-right direction. One side guide **17** and the other side guide **17** can be moved in tandem such that the side guides approach or are apart from each other by a known interlocking mechanism (not depicted) comprising, for example, racks facing each other and a pinion meshing with the racks.

(39) As described above, the feed tray **1** has a first accommodating part **8** for accommodating the rolled body R and a second accommodating part **9** for accommodating a plurality of cut papers Kp in a stacked state. The first accommodating part **8** and the second accommodating part **9** are arranged along the feeding direction. The second accommodating part **9** is located downstream in the feeding direction of the first accommodating part **8**.

(40) The first accommodating part **8** accommodates the rolled body R such that an axial direction of the rolled body R coincides with the left-right directions. As depicted in FIG. **1**, the first accommodating part **8** has two support stands **81** and **82**, and two rollers **81a** and **82a**. The two support stands **81** and **82** are spaced apart from each other in the front-rear direction. The support stand **81** is located more rearward than the support stand **82**. Each of the two support stands **81** and **82** extends along the left-right direction. The two rollers **81a** and **82a** are spaced apart from each other in the front-rear direction. The roller **81a** is mounted on the support stand **81** to be rotatable around a rotation axis extending along the left-right direction. The roller **82a** is mounted on the support stand **82** to be rotatable around a rotation axis extending along the left-right direction. The rollers **81a** and **82a** support the rolled body R from below in a state where the rollers **81a** and **82a** contact an outer circumferential surface of a lower portion of the rolled body R.

(41) The rolled paper Rp drawn from the rolled body R accommodated in the first accommodating part **8** is drawn from a front side of the support stand **81** to a rear side of the support stand **81**, through a gap between a lower surface of the support stand **81** and the bottom surface **11a**.

(42) The second accommodating part **9** has a support plate **91** for supporting, from below, the plurality of cut papers Kp in a stacked state. A rotation axis **91a** (see FIG. **1**) extending along the left-right direction is provided at a front end of the support plate **91**. The support plate **91** is rotatable around the rotation axis **91a**. The support plate **91** extends toward downstream side in the feed direction from the rotation axis **91a** to a vicinity of a downstream end of the bottom wall **11** in the feed direction.

(43) As depicted in FIG. **1**, a feed path I of the rolled paper Rp, which has been unrolled from the rolled body R accommodated in the first compartment **8**, is defined by the bottom wall **11** of the feed tray **1** and the support plate **91**. The support plate **91** partitions an inner space of the feed tray **1** into a lower space as the feed path I for the rolled paper Rp and an upper space as the second accommodating part **9** for the plurality of cut papers Kp. Namely, the support plate **91** corresponds to the “partitioning member” of the present teaching.

(44) The bottom surface **11a** of the feed tray **1** supports the rolled paper Rp passing through the feed path I. As depicted in FIG. **2**, the bottom surface **11a** has two rollers **18** arranged side by side along the left-right direction. The bottom surface **11a** and outer circumferential surfaces of the rollers **18** contact, from below, the rolled paper Rp passing through the feed path I. Namely, the bottom surface **11a** and the outer circumferential surfaces of the rollers **18** correspond to the

“contact surface” of the present teaching. The two rollers **18** are located in a central part in the left-right direction in a vicinity of a downstream end in the feed direction of the bottom surface **11a**. The two rollers **2a** configuring the feed roller **2** are contactable with the two rollers **18**, respectively.

(45) As depicted in FIG. 2, the bottom surface **11a** of the feed tray **1** is formed with two grooves **19**. The two grooves **19** are located on both sides of the two rollers **18** with respect to the left-right direction. Each groove **19** extends along the feed direction. A recess **19a** is formed at an upstream end in the feed direction in the bottom of each groove **19**.

(46) As depicted in FIGS. 2 and 3, a notch **91b** is formed at a downstream end in the feed direction of the support plate **91**. The notch **91b** is located in a central part in the left-right direction of the support plate **91**. The notch **91b** is located on a movement path of the feed roller **2**, which moves from a position where the feed roller **2** contacts with the rollers **18** to a position where the feed roller **2** is separated from the rollers **18**. The two rollers **18** and the two grooves **19** on the bottom surface **11a** of the feed tray **1** are located inside the notch **91b** of the support plate **91** in a top view. A flap **51** is attached to an inside of the notch **91b** of the support plate **91**, via a link member **61**.

(47) As depicted in FIG. 2, the support plate **91** has support portions **91c** for supporting the flap **51** attached to the notch **91b**. The support portions **91c** are located at both edges in the left-right direction in the notch **91b**. The support portions **91c** support, from below, the flap **51** in a second position (see FIG. 3), as described below.

(48) The flap **51** is a plate-shaped member. The flap **51** corresponds to the “support member” of the present teaching. One side of the flap **51** forms a support surface **51a** for supporting the plurality of cut papers **Kp** accommodated in the second accommodating part **9**.

(49) The flap **51** is configured to rotate about an axis **71** (corresponding to the “first axis” of the present teaching: see FIGS. 4A and 4B) extending in one direction in a plane parallel to the support surface **51a**. By rotating about the axis **71**, the flap **51** can be moved between a first position (see FIGS. 2 and 4A) and a second position (see FIGS. 3 and 4B). The first position is a position where the flap **51** is out of the movement path of the feed roller **2**. The second position is a position where the flap **51** is on the movement path of the feed roller **2** with the support surface **51a** facing upward. In a state that the flap **51** is in the second position, the flap **51** extends to a downstream side in the feed direction from the movement path of the feed roller **2**. As depicted in FIG. 4B, when the flap **51** is in the second position, the flap **51** covers the rollers **18** provided on the bottom surface **11a**.

(50) When recording an image on the rolled paper **Rp** unrolled from the rolled body **R** accommodated in the first accommodating part **8**, the flap **51** is positioned in the first position. When recording an image on a sheet of the cut paper **Kp** accommodated in the second accommodating unit **9**, the flap **51** is positioned in the second position.

(51) As depicted in FIG. 5, the link member **61** is located between the flap **51** and the support plate **91** in the feed direction, in the state that the flap **51** is in the second position. The link member **61** is a plate-shaped member which is long in the left-right direction. One end in a short direction of the link member **61** is attached to the support plate **91**, and the other end in the short direction of the link member **61** is attached to the flap **51**. A length along the left-right direction of the link member **61** is shorter than a length along the left-right direction of the flap **51**. The link member **61** has a protruding portion **62** protruding to the right at a right end of the other end in the short direction.

(52) As depicted in FIG. 4B, the link member **61** is attached to the support plate **91**, such that an upstream end in the feed direction of the link member **61** is rotatable around an axis **72**

(corresponding to the “second axis” of the present teaching) extending along the left-right direction in the state that the flap **51** is in the second position.

(53) As depicted in FIG. 4B, in the state that the flap **51** is in the second position, the shaft **71** (corresponding to the “first axis” of the present teaching) described above is provided at a base end of the flap **51**, which is an upstream end in the feed direction of the flap **51**. In the state that the flap

51 is in the second position, the flap **51** is attached to a downstream end in the feed direction of the link member **61** to be rotatable around the axis **71**. As depicted in FIG. 5, a central part in the left-right direction of the flap **51** is attached to the link member **61**.

(54) The flap **51** moves from the second position (see FIG. 4B) to the first position (see FIG. 4A). With the flap **51** in the second position, when the link member **61** rotates counterclockwise around the axis **72** and the flap **51** rotates counterclockwise around the axis **71**. When the flap **51** is in the first position, the flap **51** is on the upper surface of the support plate **91** with the support surface **51a** facing downward.

(55) As depicted in FIG. 6, when the flap **51** moves from the second position to the first position, the protruding portion **62** of the link member **61** interferes with the support surface **51a** of the flap **51**. This prevents an angle between the bottom surface **11a** of the feed tray **1** and the link member **61** from being 90 degrees or more. In other words, the protruding portion **62** of the link member **61** functions as the “regulating part” of the present teaching.

(56) As depicted in FIG. 3, the support surface **51a** of the flap **51** has a base pad **92**. When the flap **51** is in the second position, a portion where the base pad **92** is provided is located on the movement path of the feed roller **2**.

(57) A coefficient of static friction of the base pad **92** is greater than a coefficient of static friction of the support surface **51a**. In other words, in the support surface **51a**, an area where the base pad **92** is not provided corresponds to the “first area” of the present teaching and the area where the base pad **92** is provided corresponds to the “second area” of the present teaching. The relationship of the coefficient of friction between the base pad **92** and the support surface **51a** may be specified not only between the coefficients of static friction as in the present embodiment, but also between coefficients of dynamic friction. It may also be specified between a static friction coefficient of either one and a dynamic friction coefficient of the other.

(58) As depicted in FIG. 2, the flap **51** has two first protrusions **52** on a surface **51b** opposite to the support surface **51a**. The two first protrusions **52** are separated from each other in the left-right direction.

(59) As depicted in FIGS. 2 and 4A, in a state that the flap **51** is in the first position, the surface **51b** faces upward. Thus, in the state that the flap **51** is in the first position, the first protrusions **52** project upward.

(60) As depicted in FIG. 7, when the flap **51** is in the second position, the face **51b** faces downward. Thus, when the flap **51** is in the second position, the two first protrusions **52** project downward. When the flap **51** is in the second position, the two first protrusions **52** are received in the recesses **19a** of the two grooves **19**, respectively. As depicted in FIG. 7, with the flap **51** in the second position, an upstream part in the feed direction of the tip of each first protrusion **52** is chamfered.

(61) As depicted in FIG. 2, the two recesses **19a** for receiving the two first protrusions **52** respectively are located between the first accommodating part **8** and the two rollers **18** arranged on the bottom surface **11a**, with respect to the feed direction. That is, when the flap **51** is in the second position, the two first protrusions **52** are positioned between the first accommodating part **8** and the two rollers **18** arranged on the bottom surface **11a**. The first protrusions **52** are positioned at positions where the first protrusions **52** prevent the rolled paper Rp unrolled from the rolled body R accommodated in the first accommodating part **8** from proceeding toward the rollers **18**.

(62) As depicted in FIG. 2, two second protrusions **53** are formed on the surface **51b** of the flap **51**. The two second protrusions **53** are spaced apart from each other in the left-right direction. The two second protrusions **53** are provided in line with the two first protrusions **52** along the feed direction. As depicted in FIG. 7, the second protrusions **53** are provided in a vicinity of an end opposite to the base end of the flap **51**, with respect to a direction perpendicular to the left-right direction. When the flap **51** is in the second position, the two second protrusions **53** are received in the two grooves **19** formed in the bottom surface **11a**. A downstream side wall in the feed direction of each groove

19 is an abutment surface **19b**, against which corresponding second protrusion **53** abut. The abutment surface **19b** faces upstream in the feed direction.

(63) The flap **51** has two openings **54**, as depicted in FIG. 2. Each opening **54** extends along the feed direction. The two openings **54** are located between the two first protrusions **52** and the two second protrusions **53** arranged in the feed direction. As depicted in FIG. 3, with the flap **51** in the second position, downstream ends in the feed direction of the openings **54** are located on both sides of the base pad **92** with respect to the left-right direction.

(64) As depicted in FIG. 8A, a rolled paper label **55**, which informs the user that the rolled paper Rp is to be set, is attached to the surface **51b** facing upward with the flap **51** in the first position. The rolled paper label **55** has an illustration of the rolled body R.

(65) As depicted in FIG. 8B, a cut paper label **56**, which informs the user that the cut papers Kp are to be set, is attached to the support surface **51a** facing upward with the flap **51** in the second position. The cut paper label **56** has an illustration of the cut paper Kp.

(66) As depicted in FIG. 8B, a mark **93** indicating an orientation of the cut paper Kp to be set, etc. is depicted on the upper surface of the support plate **91**. As depicted in FIG. 8A, when the flap **51** is in the first position, the mark **93** is covered by the flap **51**.

(67) A rolled paper setting procedure label **94** and a cut paper setting procedure label **95** are attached to the upper surface of the support plate **91**. The rolled paper setting procedure label **94** informs the user of a procedure for setting the rolled paper Rp and the cut paper setting procedure label **95** informs the user of a procedure for setting the cut papers Kp. As depicted in FIG. 9, the rolled paper setting procedure label **94** has an illustration illustrating the procedure for setting rolled paper Rp. The cut paper setting procedure label **95** has an illustration illustrating the procedure for setting cut papers Kp.

(68) A color of a background of the rolled paper label **55** and a color of a background of the rolled paper setting procedure label **94** are the same color. A color of a background of the cut paper label **56** and a color of a background of the cut paper setting procedure label **95** are the same color. The color of the background of the rolled paper label **55** and the rolled paper setting procedure label **94** and the color of the background of cut paper label **56** and the cut paper setting procedure label **95** are different from each other. For example, the background color of the rolled paper label **55** and the rolled paper setting procedure label **94** is blue, while the background color of the cut paper label **56** and the cut paper setting procedure label **95** is green. This allows the user to easily distinguish whether the label is informing matters related to the rolled paper Rp or the cut paper Kp.

(69) (Configuration of Paper Sheet Presence/Absence Detecting Unit **20**)

(70) Referring to FIGS. 10A and 10B, a configuration of the paper sheet presence/absence detecting unit **20** is described. The paper sheet presence/absence detecting unit **20** includes an actuator **21** and a sensor **22**. Actuator **21** is supported by the arm **2b** to be rotatable around a rotation axis **21a**. An axial direction of the rotation axis **21a** is parallel to the left-right direction. The sensor **22** is a sensor that detects the actuator **21**. The sensor **22** is attached to the arm **2b**. The sensor **22** is, for example, a photo sensor having a light emitting element and a light receiving element (not depicted). The actuator **21** is biased by a force member (not depicted) in a clockwise direction in FIGS. 10A and 10B. As depicted in FIG. 10A, when no external force is applied to the actuator **21**, one end of the actuator **21** (the right end in FIG. 10A) is positioned below the lower end of feed roller **2**. At this time, the other end of the actuator **21** (left end in FIG. 10A) does not block an optical path of the sensor **22**.

(71) As depicted in FIG. 10A, when the feed roller **2** contacts the rollers **18** in a state that the rolled paper Rp is not in a feed position (in a state that the rolled paper Rp is not in a position, in the bottom surface **11a**, where the rollers **18** are not provided) one end of the actuator **21** is located inside the groove **19** and below the lower end of the feed roller **2**. Namely, the other end of the actuator **21** does not block the optical path of the sensor **22**.

(72) On the other hand, when the feed roller **2** and the rollers **18** nip the rolled paper Rp

therebetween in a state that the rolled paper Rp is in the feed position (in a state that the rolled paper Rp is in the position, in the bottom surface **11a**, where the rollers **18** are provided), the one end of the actuator **21** contacts the rolled paper Rp. As a result, the one end of the actuator **21** is pushed upward by the rolled paper Rp, and the actuator **21** rotates counterclockwise around the rotation axis **21a**, as depicted in FIG. **10B**. At this time, the other end of the actuator **21** blocks the optical path of the sensor **22**. This allows the paper sheet presence/absence detecting unit **20** to detect that the paper sheet P is positioned at the feed position of the feed roller **2**.

(73) When the feed roller **2** contacts the base pad **92** in a state that the cut paper Kp is not in the feed position (in a state that the cut paper Kp is not in a position, in the support surface **51a**, where the base pad **92** is provided), as depicted in FIG. **10A**, one end of the actuator **21** is located inside the opening **54** and below the lower end of the feed roller **2**. Namely, the other end of the actuator **21** does not block the optical path of the sensor **22**.

(74) On the other hand, when the feed roller **2** contacts the cut paper Kp in a state that the cut paper Kp is in the feed position (in a state that the cut paper Kp is in the position, in the support surface **51a**, where the base pad **92** is provided), the one end of the actuator **21** also contacts the cut paper Kp. As a result, the one end of the actuator **21** is pushed upward by the cut paper Kp, and the actuator **21** is rotated counterclockwise around the rotation axis **21a**, as depicted in FIG. **10B**. At this time, the other end of the actuator **21** blocks the optical path of the sensor **22**. This allows the paper sheet presence/absence detecting unit **20** to detect that the paper sheet P is positioned at the feed position of the feed roller **2**.

(75) (Retraction Mechanism of Arm **2b**)

(76) A retraction mechanism of feed roller **2** will be described with reference to FIG. **11**. As depicted in FIGS. **1** and **11**, the arm **2b** of the feed roller **2** has an extending portion **2d** extending along the left-right direction. A roller **2e** is attached to an end in the left-right direction of the extending portion **2d** to be rotatable around a rotation axis extending along the left-right direction. As depicted in FIG. **1**, when the feed tray **1** is inside the housing **100a**, the feed roller **2** is in the feed position. At this time, the roller **2e** is in a notch **13a** (notch **12a**) formed in the side wall **13** (side wall **12**).

(77) As the feed tray **1** is pulled out from inside the housing **100a** and moves frontward, as depicted in FIG. **11**, the roller **2e** rolls to an upper surface of the side wall **13** (side wall **12**) over the notch **13a** (notch **12a**). As a result, the arm **2b** pivots upward around the spindle **2c**, and the arm **2b** moves to the retracted position.

(78) (Procedure for Setting Rolled Paper Rp and Cut Paper Kp)

(79) When recording images on the rolled paper Rp, the rolled body R is accommodated in the first accommodating part **8** of the feed tray **1**. At this time, the flap **51** is set to the first position. Then, the rolled paper Rp unrolled from the rolled body R is fed to the downstream side in the feed direction from the rollers **18**. In this state, the feed tray **1** is moved from outside of the housing **100a** to inside of the housing **100a**. As a result, the feed roller **2** and the rollers **18** nip the rolled paper Rp and make it possible to feed the rolled paper Rp.

(80) When recording images on the cut papers Kp, the flap **51** is set to the second position and the cut papers Kp are accommodated in the second accommodating part of feed tray **1** in a stacked state. As a result, the support surface **51a** of the flap **51** supports the cut papers Kp. In this state, the feed tray **1** is moved from the outside of the housing **100a** to the inside of the housing **100a**. As a result, the feed roller **2** contacts the uppermost cut paper Kp of the cut papers Kp and makes it possible to feed the cut paper Kp.

Features of the Embodiment

(81) As described above, the printer **100** in this embodiment includes the feed tray **1**, the feed roller **2** which can feed the paper sheets P accommodated in the feed tray **1**, and a head **6** which records images on the paper sheet P fed by the feed roller **2**. The feed tray **1** has the first accommodating part **8** for accommodating the rolled body R in which the paper sheet P is rolled, the second

accommodating part **9** for accommodating the cut papers Kp in a stacked state, the rollers **18** which contact, from below, the paper sheet P unrolled from the rolled body, and the flap **51** having the support surface **51a** for supporting the cut papers Kp accommodated in the second accommodating part **9**. The feed roller **2** is configured to move between the position where the feed roller **2** contacts with the rollers **18** and the position where the feed roller **2** is away from the rollers **18**. The flap **51** is configured to be movable between the first position where the flap **51** is retracted from (out of) the movement path of the feed roller **2** and the second position where the flap **51** is positioned on the movement path of the feed roller **2** with the support surface **51a** facing upward.

(82) With the above configuration, when feeding the rolled paper Rp from the first accommodating part **8** of the feed tray **1**, the flap **51** is set to the **1** position, and the rolled paper Rp in contact with the rollers **18** can be fed by the feed roller **2**. When feeding the cut paper Kp from the second accommodating part **9** of the feed tray **1**, the flap **51** is set to the second position, and the cut paper Kp supported by the support surface **51a** can be fed by the feed roller **2**. Namely, the rolled paper Rp unrolled from the rolled body R in the first accommodating part **8** and the cut paper Kp in the second accommodating part **9** can be fed by the single feed roller **2** along the same route.

Therefore, the size of the apparatus can be reduced.

(83) In the printer **100** of the above embodiment, it is possible to nip the rolled paper Rp between the rollers **18** and the feed roller **2** and to convey the rolled paper Rp smoothly.

(84) In the printer **100** of the embodiment described above, the support surface **51a** of the flap **51** has the base pad **92**. The coefficient of friction of the base pad **92** is greater than the coefficient of friction of the support surface **51a**. When the flap **51** is in the second position, the portion, of the flap **51**, where the base pad **92** is provided is located on the movement path of the feed roller **2**. By providing the base pad **92** on the support surface **51a**, it is possible to prevent a lowermost cut paper Kp among the cut paper Kp in the stacked state from moving, and to prevent feeding the cut papers Kp in a state of being overlapped with each other. In particular, when there are two sheets of the cut papers Kp on the support surface **51a**, it is possible to prevent effectively feeding the two sheets of the cut papers Kp in the state of being overlapped with each other. When the area where the base pad **92** is provided is out of the movement path of the feed roller **2**, there is a possibility that the cut papers Kp are fed in a state of being overlapped with each other (overlapping feed) or that the cut papers Kp may not be fed (blank feed). In this configuration, because the part where the base pad **92** is provided is located on the movement path of the feed roller **2**, it is possible to prevent the overlapping feed and the blank feed, certainly.

(85) In the printer **100** of the embodiment described above, when the flap **51** is in the second position, the flap **51** covers the contact portion, of the bottom surface **11a**, where the feed roller **2** contacts. According to this configuration, even if the rolled paper Rp is accidentally fed into the feed path I when the flap **51** is in the second position, the feed roller **2** cannot feed the rolled paper Rp on the bottom surface **11a** because the flap **51** covers the part, of the bottom surface **11a**, where the feeding roller **2** contacts. Therefore, the rolled paper Rp cannot be fed out from the feed tray **1**.

(86) In the printer **100** of the embodiment described above, when the paper sheet presence/absence detecting unit **20** detects that the paper sheet P is placed on the feed position, controller **10** controls the feed roller **2** to feed the paper sheet P from the feed tray **1**. When the flap **51** is in the second position, even if the cut paper Kp is not supported on the support surface **51a** of the flap **51** and the rolled paper Rp is accidentally fed into the feed path I, the feed roller **2** contacts the support surface **51a** of the flap **51**. At this time, the paper sheet presence/absence detecting unit **20** does not detect that the paper sheet P is placed on the feed position. Therefore, the rolled paper Rp is not fed out from the feed tray **1** by feed roller **2**.

(87) In the feed tray **1** of the embodiment described above, the surface **51b** opposite the support surface **51a** of the flap **51** has the first protrusions **52**. When the flap **51** is in the second position, the first protrusions **52** prevent the rolled paper Rp unrolled from the rolled body R accommodated in the first accommodating part **8** from proceeding toward the rollers **18** between the first

accommodating part **8** and the rollers **18**. When flap **51** is in the second position, even if the rolled paper Rp unrolled from the rolled body R accommodated in the first accommodating part **8** is accidentally Rp fed toward the rollers **18**, the first protrusions **52** prevents the rolled paper Rp from being conveyed to the rollers **18**. This prevents the rolled paper Rp from being accidentally fed out from the feed tray **1**.

(88) In the feed tray **1** of the embodiment described above, when the flap **51** is in the first position, the surface **51b** having the first protrusions **52** faces upward. Therefore, with the flap **51** in the first position, if the cut papers Kp are accidentally placed on the surface **51b** opposite to the support surface **51a** of the flap **51**, the first protrusions **52** will lift the cut papers Kp. Therefore, the user can easily notice the setting error.

(89) In the feed tray **1** of the embodiment described above, the bottom surface **11a** has the recesses **19a** for receiving the first protrusions **52** of the flap **51** in the second position, and the tip end of each of the first protrusions **52** is chamfered. Therefore, when the flap **51** is moved from the first position to the second position, it is possible to prevent the first protrusions **52** from riding up on the bottom surface **11a** of the feed tray **1**.

(90) In the feed tray **1** of the embodiment described above, the surface **51b** opposite to the support surface **51a** of the flap **51** has the second protrusions **53**. The bottom wall **11** of the main body **16** has the abutment surface **19b** facing upstream in the feed direction. The second protrusions **53** abut against the abutment surface **19b** when the flap **51** is in the second position. When the cut paper Kp supported by the support surface **51a** of the flap **51** is fed by the feed roller **2** from the feed tray **1**, even if the force toward the downstream side of the feed direction is applied to the flap **51**, the second protrusions **53** of the flap **51** abuts against the abutment surface **19b** of the main body **16**. This prevents a large force from being applied to the axes **71**, **72**, etc.

(91) The feed tray **1** of the embodiment described above has the link member **61** via which the flap **51** is attached to the support plate **91**. The link member **61** is positioned between the flap **51** and the support plate **91** in the feed direction when the flap **51** is in the second position. The upstream end in the feed direction of the link member **61** is attached to the support plate **91**, such that the link member **61** can rotate about the axis **72** extending along the left-right direction. When the flap **51** is in the second position, the flap **51** is attached to the downstream end in the feed direction of the link member **61**, such that the flap **51** is rotatable about the axis **71** extending along the left-right direction. The link member **61** is provided with the protruding portion **62**. When the flap **51** rotates and moves from the second position to the first position, the protruding portion **62** interferes with the support plate **91**, and prevent the angle between the bottom surface **11a** of the feed tray **1** and the link member **61** from being 90 degrees or more. Therefore, when moving the flap **51** from the second position to the first position, the flap **51** can be reliably placed on the top surface of the support plate **91**. In addition, since the angle between the bottom surface **11a** of the feed tray **1** and the link member **61** is less than 90 degrees, the flap **51** can be placed in an appropriate position.

(92) In the feed tray **1** of the embodiment described above, the link member **61** is the plate-shaped member which is long in the left-right direction. One end in the short direction of the link member **61** is attached to the support plate **91**. The other end in the short direction of the link member **61** is attached to the central part in the left-right direction of the flap **51**. The right end at the other end in the short direction of the link member **61** has the protruding portion **62** protruding to the right. Thus, the length of the other end in the short direction of the link member **61** is longer than the length in the short direction of the one end of the link member **61**. Therefore, when attaching the link member **61** to the support plate **91** and the flap **51**, it is possible to prevent the one end in the short direction from being attached to the flap **51** and the other end in the short direction from being attached to the support plate **91**. Also, when attaching the link member **61** to the support plate **91** and the flap **51**, if the front and back surfaces of the link member **61** are opposite each other, the protruding portion **62** is located at the left end. Therefore, the position of the flap **51** attached to the other end of the link member **61** is shifted to the left. Thus, the user can easily notice that the front

and back surfaces of the link member **61** are opposite to each other.

(93) In the feed tray **1** of the embodiment described above, the notch **91b** is formed in the downstream end in the feed direction of the support plate **91**, and the flap **51** is attached to the inside of the notch **91b** of the support plate **91**. Since the support plate **91** is located on both sides in the left-right direction of the flap **51**, the lift of the rolled paper Rp supported by the bottom surface **11a** of the feeding tray **1** can be suppressed by the support plate **91**.

(94) The feed tray **1** of the embodiment described above is provided with the separation strip **31**. The separation strip **31** contacts the cut paper Kp fed from the second accommodating part **9** by the feed roller **2** and separates the cut paper Kp which the feed roller **2** contacts from the other cut papers Kp. When the flap **51** is in the second position, the flap **51** extends downstream in the feed direction from the movement path of the feed roller **2**. Therefore, the support surface **51a** of the flap **51** supports the cut paper Kp and maintain the posture of the cut paper Kp even downstream in the feed direction from the movement path of the feed roller **2**. Therefore, the cut paper Kp can be securely separated by the separation strip **31**. A curl of the rolled paper Rp supported by the bottom surface **11a** can be suppressed by the flap **51** to the feed position.

(95) In the feed tray **1** of the embodiment described above, the support plate **91** has the support portion **91c** which supports the flap **51** in the second position from below. The support plate **91** is configured to be rotatable around the rotation axis **91a** extending along the left-right direction. Accordingly, when the support plate **91** is rotated around the rotation axis **91a**, the flap **51** supported by the support portion **91c** also rotates together with the support plate **91**.

(96) The above description of the embodiment of the present teaching is based on the drawings, but the specific configurations should not be considered limited to the embodiment. The scope of the teaching is depicted by the claims, not by the description of the above embodiment, and further includes all modifications within the meaning and scope equivalent to the claims.

(97) In the embodiment described above, the bottom surface **11a** of the feed tray **1** is provided with rollers **18** at the portion where the feed roller **2** contacts. However, the rollers **18** may not be provided, and the feed roller **2** may contact the bottom surface **11a** directly. In place of the rollers **18**, a base pad or a roller having a coefficient of friction smaller than that of the bottom surface **11a** may be provided.

(98) In the embodiment described above, the support surface **51a** of the flap **51** is provided with the base pad **92** having the coefficient of friction greater than the coefficient of friction of the support surface **51a**. The base pad **92** is located on the movement path of the feed roller **2** when the flap **51** is in the second position. However, the present teaching is not limited to this, and the base pad **92** may be provided at a position out of the movement path of the feed roller **2** when the flap **51** is in the second position. Furthermore, instead of the base pad **92**, an area with a higher coefficient of friction than other area of the support surface **51a** may be provided by processing the support surface **51a**. The support surface **51a** may not be provided with areas having different coefficients of friction.

(99) In the embodiment described above, when the paper sheet presence/absence detecting unit **20** detects that paper sheet P is placed in the feed position, the controller **10** controls feed roller **2** to feed the paper sheet P from feed tray **1**. However, the present teaching is not limited to this. That is, the paper sheet presence/absence detecting unit **20** may not be provided.

(100) In the embodiment described above, the flap **51** is provided as a support member with a support surface for supporting the cut papers Kp accommodated in the second accommodating part **9**. The flap **51** rotates around the axis **71** extending along the left-right direction to move between the first position in which the flap **51** is evacuated from (out of) the movement path of the feed roller **2** and the second position in which the flap **51** is on the movement path of the feed roller **2** with the support surface **51a** facing upward. However, the present teaching is not limited to this. For example, in the feed tray **201** depicted in FIGS. **12a** and **12b**, the support member **51** is configured to slide along the feed direction. When the support member **51** is in the first position

where the support member **51** is evacuated from the movement path of the feed roller **2** as depicted in FIG. **12A**, and when the support member **51** is in the second position on the movement path of the feed roller **2** as depicted in FIG. **12B**, the support surface **51a** faces upward. Furthermore, although not depicted in the drawings, the support member may be configured to be removable from the feed tray **1**. In this case, by removing the support member from the feed tray **1**, the support member can be evacuated from the movement path of the feed roller **2**.

(101) In the embodiment described above, the surface **51b** of the flap **51** is provided with the first protrusions **52**. The first protrusions **52** are received in the recesses **19a** formed in the bottom surface **11a** of the feed tray **1**, when the flap **51** is in the second position. However, the present teaching is not limited to this. For example, the recesses **19a** may not be formed in the bottom surface **11a** of the feed tray **1**, and the first protrusions **52** may contact the bottom surface **11a** when the flap **51** is in the second position.

(102) Although the tips of the first protrusions **52** are chamfered in the embodiment described above, the tips of the first protrusions **52** may not be chamfered.

(103) In the embodiment described above, the surface **51b** of the flap **51** has second protrusions **53** which contact the abutment surfaces **19b** of the main body **16** when the flap **51** is in the second position. However, the second protrusions **53** and the abutment surface **19b** may not be provided.

(104) In the embodiment described above, the flap **51** is attached to the support plate **91** via the link member **61**. However, the flap **51** may be directly attached to the supporting plate **91** without any other member.

(105) In the embodiment described above, the link member **61** has the protruding portion **62**. However, the link member **61** may not have the protruding portion **62**.

(106) In the embodiment described above, a notch **91b** is formed at the downstream end in the feed direction of the support plate **91**, and the flap **51** is attached to the inside of the notch **91b** of the support plate **91**. However, the present teaching is not limited to this. For example, the notch **91b** may not be formed in the support plate **91**, and the flap **51** may be attached to the downstream end in the feed direction of the support plate **91**.

(107) In the embodiment described above, when the flap **51** is in the second position, the flap **51** extends further downstream in the feed direction than the movement path of the feed roller **2**. However, the present teaching is not limited to this. For example, the downstream end in the feed direction of the flap **51** in the second position may be located on the movement path of the feed roller **2**.

(108) In the embodiment described above, the support plate **91** has the support portion **91c** supporting the flap **51** from below, and the support plate **91** is configured to be rotatable around the rotation axis **91a** extending along the left-right direction. However, the present teaching is not limited to this. For example, the support plate **91** may not have the support portion **91c**. Further, the support plate **91** may not be rotatable around the rotation axis **91a**.

(109) The present teaching can be applied to all image recording apparatuses provided with a feed tray which can accommodate a rolled body **R** and cut papers **Kp**. In other words, the present teaching can be applied not only to an inkjet printer provided with a head, which ejects ink from nozzles, as a recording unit, but also to a laser-type printer where an electrostatic latent image is formed by exposing a photosensitive material with a laser, or to an electrophotographic printer equipped with an LED-type recording section where an electrostatic latent image is formed by exposing a photosensitive material with a LED. The sheet media is not limited to the paper, but can also be a cloth or other resin material such as a film and the like, as long as having a sheet form.

Claims

1. An image recording apparatus, comprising: a feed tray; a feed roller configured to feed a sheet medium accommodated in the feed tray; and a recording unit configured to record an image on the

sheet medium fed by the feed roller, wherein the feed tray includes: a first accommodating part configured to accommodate a rolled body formed by rolling the sheet medium; a second accommodating part configured to accommodate a plurality of sheet media in a stacked state, each of the sheet media being the sheet medium; a contact surface configured to contact, from below, the sheet medium unrolled from the rolled body accommodated in the first accommodating part; and a support member having a support surface configured to support the plurality of sheet media in the stacked state, which is accommodated in the second accommodating part, in a state where the second accommodating part is in a posture to accommodate the plurality of sheet media in the stacked state, a position of the second accommodating part relative to the first accommodating part is fixed, the feed roller is configured to be movable between a contact position at which the feed roller contacts the contact surface and a separation position at which the feed roller is away from the contact surface, and the support member is configured to be movable between a first position at which the support member is outside a movement path of the feed roller and a second position at which the support member is on the movement path of the feed roller with the support surface facing upward.

2. The image recording apparatus according to claim 1, wherein the feed tray further includes a roller having the contact surface.

3. An image recording apparatus comprising: a feed tray; a feed roller configured to feed a sheet medium accommodated in the feed tray; and a recording unit configured to record an image on the sheet medium fed by the feed roller, wherein the feed tray includes: a first accommodating part configured to accommodate a rolled body formed by rolling the sheet medium; a second accommodating part configured to accommodate a plurality of sheet media in a stacked state, each of the sheet media being the sheet medium; a contact surface configured to contact, from below, the sheet medium unrolled from the rolled body accommodated in the first accommodating part; a support member having a support surface configured to support the sheet media accommodated in the second accommodating part, and a roller having the contact surface, the feed roller is configured to be movable between a contact position at which the feed roller contacts the contact surface and a separation position at which the feed roller is away from the contact surface, the support member is configured to be movable between a first position at which the support member is outside a movement path of the feed roller and a second position at which the support member is on the movement path of the feed roller with the support surface facing upward, the support surface includes a first area and a second area having a coefficient of friction higher than a coefficient of friction of the first area, and in a state that the support member is positioned at the second position, the second area is positioned on the movement path of the feed roller.

4. The image recording apparatus according to claim 1, wherein in the second position, the support member covers a contact part, of the contact surface, at which the feed roller contacts the contact surface.

5. The image recording apparatus according to claim 1, further comprising: a controller; and a detecting unit configured to detect whether the sheet medium is placed at a feed position at which the sheet medium is fed by the feed roller, wherein in a case that the detecting unit detects that the sheet medium is placed at the feed position, the controller is configured to control the feed roller to feed the sheet medium from the feed tray.

6. An image recording apparatus comprising: a feed tray; a feed roller configured to feed a sheet medium accommodated in the feed tray; and a recording unit configured to record an image on the sheet medium fed by the feed roller, wherein the feed tray includes: a first accommodating part configured to accommodate a rolled body formed by rolling the sheet medium; a second accommodating part configured to accommodate a plurality of sheet media in a stacked state, each of the sheet media being the sheet medium; a contact surface configured to contact, from below, the sheet medium unrolled from the rolled body accommodated in the first accommodating part; and a support member having a support surface configured to support the sheet media accommodated in

the second accommodating part, the feed roller is configured to be movable between a contact position at which the feed roller contacts the contact surface and a separation position at which the feed roller is away from the contact surface, the support member is configured to be movable between a first position at which the support member is outside a movement path of the feed roller and a second position at which the support member is on the movement path of the feed roller with the support surface facing upward, and the support member is a plate member with one side as the support surface, and moves between the first position and the second position by rotating around a first axis extending in one direction in a plane parallel to the support surface.

7. The image recording apparatus according to claim 6, wherein the feed tray further includes a partitioning member configured to partition a space in the feed tray into an upper space and a lower space, the first axis is parallel to a horizontal direction and extends in a first direction orthogonal to a feed direction of the sheet medium by the feed roller, the support member has a base end, at which the first axis is located, at an upstream end in the feed direction in a state of the support member being in the second position, and the base end of the support member is attached to a downstream end in the feed direction of the partitioning member.

8. The image recording apparatus according to claim 7, wherein the support member has a first protrusion formed on the other side opposite to the one side, and in the state of the support member being in the second position, the first protrusion is arranged at a position between the first accommodating part and a contact part, of the contact surface, at which the feed roller contacts the contact surface, such that the first protrusion prevents the sheet medium unrolled from the rolled body from proceeding toward the contact part.

9. The image recording apparatus according to claim 8, wherein the other side of the support member faces upward in a state of the support member being in the first position.

10. The image recording apparatus according to claim 8, wherein the contact surface has a recess configured to receive the first protrusion of the support member in the second position, and a tip end of the first protrusion is chamfered.

11. The image recording apparatus according to claim 7, wherein the feed tray further includes a main body, the support member has a second protrusion formed on the other side opposite to the one side, and the main body has an abutment surface which faces upstream side in the feed direction such that the second protrusion abuts against the abutment surface in the state of the support member being in the second position.

12. The image recording apparatus according to claim 7, wherein the support member is on an upper surface of the partitioning member in a state of the support member being in the first position.

13. The image recording apparatus according to claim 12, wherein the feed tray further includes a link member via which the support member is attached to the partitioning member, in the state of the support member being in the second position, the link member is arranged between the support member and the partitioning member in the feed direction, and an upstream end in the feed direction of the link member is attached to the partitioning member such that the link member rotates around a second axis extending in the first direction, in the state of the support member being in the second position, the support member is attached to a downstream end in the feed direction of the link member such that the support member rotates around the first axis, and the link member includes a regulating part configured to interfere with the support member to prevent an angle between the contact surface and the link member from being 90 degrees or more under a condition that the support member moves from the second position to the first position by rotating.

14. The image recording apparatus according to claim 13, wherein the link member is a plate member long in the first direction, one end in a short direction of the link member is attached to the partitioning member and the other end in the short direction of the link member is attached to a central part in the first direction of the support member, and the link member has a protruding portion protruding in the first direction, at one end in the first direction of the other end in the short

direction.

15. The image recording apparatus according to claim 7, wherein the partitioning member is formed with a notch at the downstream end in the feed direction, and the support member is attached to an inside of the notch of the partitioning member.

16. The image recording apparatus according to claim 7, further comprising a separation strip configured to contact the sheet medium fed from the second accommodating part by the feed roller to separate the sheet medium contacting the feed roller from other sheet media accommodated in the second accommodating part, wherein in the state of the support member being in the second position, the support member extends to a downstream side in the feed direction with respect to the movement path of the feed roller.

17. The image recording apparatus according to claim 7, wherein the partitioning member has a support portion configured to support the member, and the partitioning member is configured to rotate around a third axis extending along the first direction.

18. A feed tray, comprising; a first accommodating part configured to accommodate a rolled body formed by rolling sheet medium; a second accommodating part configured to accommodate a plurality of sheet media in a stacked state, each of the sheet media being the sheet medium; a contact surface configured to contact, from below, the sheet medium unrolled from the rolled body accommodated in the first accommodating part; and a support member having a support surface configured to support the plurality of sheet media in the stacked state, which is accommodated in the second accommodating part, wherein in a state where the second accommodating part is in a posture to accommodate the plurality of sheet media in the stacked state, a position of the second accommodating part relative to the first accommodating part is fixed, and the support member is configured to be movable between a cover position at which the support member covers a part of the contact surface with the support surface facing upward and a separation position at which the support member is farther from the part of the contact surface as compared with the cover position.

19. The image recording apparatus according to claim 1, further comprising a housing, wherein during recording by the recording unit, the first accommodating part is located inside the housing.
